NEGATIVE EXT VANISHING AND THE HOM EXACT SEQUENCE ALONG AN FPPF HYPERCOVERING

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the cohomological-descent input is card L46153 (lemma-coh-descent-qcoh-unbounded-for-fppf-hypercovering); the nullity lemma used is the modules twin of card L46186 (lemma-nullity-cartesian-objects-derived); the assumed componentwise vanishing includes the i < 0 cases needed by the cited lemmas; no verification is omitted.

Editorial note. The label lemma-fppf-neg-ext-zero-hom names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The simplicial systems of the derived category attached to La^*K and La^*E. Same chapter, Definition \label{definition-cartesian-derived-modules} and \label{lemma-cartesian-objects-derived-modules}. Proof lines 7844-7853: K_n = La_n^*K and E_n = La_n^*E are the simplicial systems of the derived category of modules associated to La^*K and La^*E, with a : U_etale -> X_etale as in the simplicial-algebraic-spaces section.
- Unbounded cohomological descent for fppf hypercoverings on QCoh. Same chapter, \label{lemma-coh-descent-qcoh-unbounded-for-fppf-hypercovering} (card L46153). Proof lines 7855-7860: Hom_{O_X}(K, E) = Hom_{O_U}(La^*K, La^*E).
- Vanishing and the Hom exact sequence for cartesian modules. Same chapter, \label{lemma-nullity-cartesian-modules-derived} and \label{lemma-hom-cartesian-modules-derived}. Proof lines 7862-7872: the first arrow is injective by the nullity lemma; its image is the kernel of the second by the Hom lemma for cartesian modules.

Source reference. Original reference 0D9N: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 3357–3357. Prerequisite pointer invoked at proof line(s) 7847; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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Source reference. Original reference 0D9P: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 3389–3389. Prerequisite pointer invoked at proof line(s) 7849; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0DHF: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 7676–7676. Prerequisite pointer invoked at proof line(s) 7853; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0D9S: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 3516–3516. Prerequisite pointer invoked at proof line(s) 7872; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0D9R: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 3493–3493. Prerequisite pointer invoked at proof line(s) 7870; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 0DE7: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 6985–6985. Prerequisite pointer invoked at proof line(s) 7851; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let X be an algebraic space over a scheme S. Let $K, E \in D_{QCoh}(\mathcal{O}_X)$. Let $a : U \to X$ be an fppf hypercovering. Assume that for all $n \geq 0$ we have*

$$\mathrm{Ext}^i_{\mathcal{O}_{U_n}}(La_n^*K, La_n^*E) = 0 \text{ for } i < 0$$

Then we have

(1) $\mathrm{Ext}^i_{\mathcal{O}_X}(K, E) = 0$ *for* $i < 0$, *and*
(2) *there is an exact sequence*

$$0 \to \mathrm{Hom}_{\mathcal{O}_X}(K, E) \to \mathrm{Hom}_{\mathcal{O}_{U_0}}(La_0^*K, La_0^*E) \to \mathrm{Hom}_{\mathcal{O}_{U_1}}(La_1^*K, La_1^*E)$$

Proof. Write $K_n = La_n^*K$ and $E_n = La_n^*E$. Then these are the simplicial systems of the derived category of modules (Definition 0D9N) associated to La^*K and La^*E (Lemma 0D9P) where $a : U_{étale} \to X_{étale}$ is as in Section 0DE7. Let us prove (2) first. By Lemma 0DHF we have

$$\mathrm{Hom}_{\mathcal{O}_X}(K, E) = \mathrm{Hom}_{\mathcal{O}_U}(La^*K, La^*E)$$

Thus the sequence looks like this:

$$0 \to \mathrm{Hom}_{\mathcal{O}_U}(La^*K, La^*E) \to \mathrm{Hom}_{\mathcal{O}_{U_0}}(K_0, E_0) \to \mathrm{Hom}_{\mathcal{O}_{U_1}}(K_1, E_1)$$

The first arrow is injective by Lemma 0D9R. The image of this arrow is the kernel of the second by Lemma 0D9S. This finishes the proof of (2). Part (1) follows by applying part (2) with $K[i]$ and E for $i > 0$. □